

Report on the PhD thesis “Enumerating score sequences and permutations by inversions and forbidden patterns”, by Atli Fannar Franklín

The PhD thesis of Atli Fannar Franklín consists of four articles on the topic of enumerative combinatorics (in a broad sense).

The first paper deals with the problem of enumerating score sequences of tournaments. Tournaments are complete directed graphs, usually studied by graph theorists mainly for what concern their structural properties. In order to find information on the number of score sequences of tournaments, Atli Fannar Franklín brings clever enumerative techniques and strategies in the realm of graph theory, thus obtaining an elegant recursion for such quantities (previously only conjectured). The proof follows from a nice factorization of the generating function of (pointed) score sequences, which reveals a masterful handling of several nontrivial combinatorial techniques.

The remaining papers all deal with permutations. The second paper provides very beautiful expressions for the generating functions of permutations with few inversions. In particular, the case of permutations whose length coincides with the number of inversions requires a very intricate argument, which has the advantage of being entirely combinatorial. The third paper presents enumerative results on indecomposable pattern-avoiding permutations in terms of the number of inversions only, for several patterns or sets of patterns. Such results show some very intriguing connections with several apparently unrelated combinatorial objects, thus witnessing the deep combinatorics that seem to be hidden in this sort of enumeration problems. Finally, the fourth paper proposes a novel approach to the study of 1324-avoiding permutations (whose enumeration is a major open problem in this area) which exploits an encoding of permutations in terms of certain walks in directed graphs. It seems plausible that this new perspective can suggest new avenues for, at least, finding more precise information on the growth rate of the class of 1324-avoiding permutations (and in fact an interesting conjecture is formulated concerning precisely this issue).

In my opinion, the PhD thesis of Atli Fannar Franklín is an extremely interesting piece of work, which describes some exciting new results on the enumeration of certain combinatorial objects and, maybe even most importantly, offers a number of ideas for further research along the lines suggested by the author. The methodologies and techniques employed by Atli Fannar Franklín are, in most cases, very elegant and not at all trivial, and show a deep understanding of the combinatorics involved in the addressed problems.

In my opinion, the present thesis has no drawbacks, neither in terms of the scientific content, nor for what concerns the writing and presentation. One very minor point that I have noticed (again, in my very personal opinion) is that, in some cases, a more careful comparison of the results obtained

with the results and techniques already reported in the literature would have added value to the thesis (which is already extremely sound in its own right). For example, in the second paper, the derivation of the formula for $S_i(x)$, $i \neq 0$, could have been equally done using Riordan arrays (notice however that this formula is the “easy” part of the paper). Another instance are the formulas for counting indecomposable permutations avoiding two or more patterns only in terms of inversions (third paper), which could have been more easily obtained by exploiting the known geometric structure of the various permutation classes (at least in some of the cases).

The above considerations are, however, entirely marginal and in no way detract from the value of the thesis. Overall, the thesis of Atli Fannar Franklín is very solid work, and I strongly recommend that he be awarded a PhD in Mathematics.

To conclude, I report just a couple of very minor remarks that have occurred to me whilst reading the thesis.

- Top of page 26: do you mean $j = n$ here (instead of $j = 0$)?
- Bottom of page 72: I am not sure that the definition of the sum of two subdiagonal subsequences is correct as it is written; if $k > 0$, you can never get a subdiagonal sequence. Maybe just remove k ? At any rate, you never use this specific way of defining the sum in the proof, so the argument is not affected by this.